

Reg No.: _____

Name: _____

APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY

First Semester B.Tech Degree Regular and Supplementary Examination December 2022 (2019 Scheme)

Course Code: EST130**Course Name: BASICS OF ELECTRICAL AND ELECTRONICS ENGINEERING
(2019 -Scheme)****PART I : BASIC ELECTRICAL ENGINEERING**

Max. Marks: 50

Duration: 90 min

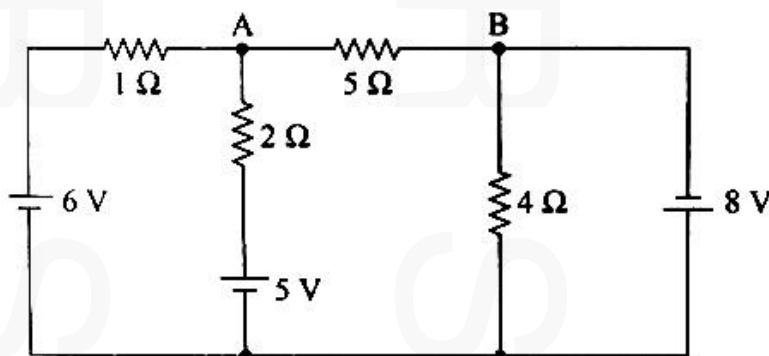
PART A*Answer all questions, each carries 4 marks*

Marks

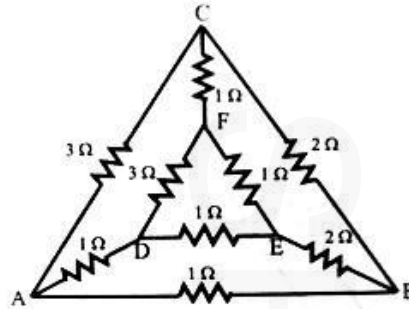
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|---|--|-----|
| 1 | State and explain Kirchhoff's laws. | (4) |
| 2 | Differentiate between ideal and real current sources with circuit representation. | (4) |
| 3 | Compare electric and magnetic circuits with circuit diagram. | (4) |
| 4 | A coil of 200 turns carries a current of 4A. The magnetic flux linkage with the coil is 0.02Wb. Calculate the self-induced emf in the coil. | (4) |
| 5 | A delta-connected load of 12Ω resistance and $16\text{-}\Omega$ reactance are connected across a 100V, 50 Hz supply. Find line current, phase current and power factor. | (4) |

PART B*Answer one full question from each module, each question carries 10 marks.***MODULE 1**

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| 6 | Calculate the current flowing through 5Ω resistor using the nodal method. | (10) |
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**OR**

- 7 Find the resistance between points A and B in network shown. (10)



MODULE 2

- 8 a) Two identical coils 1 and 2 are wound on the same magnetic core. Current in coil 1, which is changing at the rate of 600 A/s, induces emf of 12 V in coil 2. Calculate the mutual inductance between the coils. If the self-inductance of each coil is 50mH, calculate the coefficient of coupling between coils. (7)

- b) Define a) MMF b) Field Strength c) Flux Density (3)

OR

- 9 a) An alternating current is given by $14.14\sin 377t$. Find the (a) rms value of current (b) Average value of current (c) frequency (d) form factor (e) peak factor (f) instantaneous value of current when $t=3\text{ms}$. (6)
- b) Explain the terms statically induced emf and dynamically induced emf. (4)

MODULE 3

- 10 A capacitor having a capacitance of $20\mu\text{F}$ is connected in series with a non-inductive resistance of 200Ω across 220V, 50 Hz supply. Calculate the following (10)
- 1) Impedance 2) Current 3) Power Factor 4) Power drawn from supply.

OR

- 11 Show that the power consumed by three identical single-phase loads connected in delta is equal to three times the power consumed when the phase loads are connected in star. (10)

PART 2 : BASIC ELECTRONICS ENGINEERING

Max. Marks: 50

Duration: 90 min

PART A

Answer all questions, each carries 4 marks

Marks

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|----|--|-----|
| 12 | Explain any four resistor specifications. | (4) |
| 13 | Compare the three transistor configurations. | (4) |
| 14 | Why does voltage gain of an RC coupled amplifier decrease at low and high frequencies? | (4) |
| 15 | Describe the working of a full wave bridge rectifier. | (4) |
| 16 | What is the basic principle of cellular communication? | (4) |

PART B

Answer one full question from each module, each question carries 10 marks.

MODULE 4

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| 17 | a. Describe the principle of operation of an NPN transistor. | (5) |
| | b. What is an inductor? How does an inductor work? | (5) |

OR

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|----|--|-----|
| 18 | a. For an NPN transistor, $\alpha=0.95$ and $I_E=10\text{mA}$. Find I_B and I_C . | (5) |
| | b. With necessary diagram, explain the V-I characteristics of PN junction diode. | (5) |

MODULE 5

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| 19 | a. Draw the block diagram of DC power supply and explain the function of each block. | (5) |
| | b. Give the circuit diagram of an RC coupled amplifier. Explain its working. | (5) |

OR

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|----|---|-----|
| 20 | a. Give the circuit diagram of a simple zener voltage regulator. Explain its working. | (5) |
| | b. With necessary block diagram, explain an electronic instrumentation system. | (5) |

MODULE 6

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| 21 | Explain the principle and block diagram of GSM. | (10) |
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OR

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| 22 | a. With necessary block diagram explain the principle of super heterodyne receiver. | (5) |
| | b. Compare amplitude and frequency modulation. | (5) |
